



AC orbit bump method of local impedance measurement

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ABSTRACT

A fast and precise technique of local impedance measurement has been developed and tested at NSLS-II. This technique is based on in-phase sine-wave (AC) excitation of four fast correctors adjacent to the vacuum chamber section, impedance of which is measured. The beam position is measured using synchronous detection. Use of the narrow-band sine-wave signal allows us to improve significantly the accuracy of the orbit bump method. Beam excitation by fast correctors results in elimination of the systematic error caused by hysteresis effect. The systematic error caused by orbit drift is also eliminated because the measured signal is not affected by the orbit motion outside the excitation frequency range. In this article, the measurement technique is described and the result of proof-of-principle experiment carried out at NSLS-II is presented.

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1. Introduction

High intensity of particle beams is crucial for effective and high-performance operations of modern storage rings, both colliders and light sources. The beam intensity is limited by collective effects of beam dynamics, i.e. perturbations of the particle beam motion depending on its intensity. The collective effects result from the interaction of a particle beam with electromagnetic fields induced in a vacuum chamber by the beam itself. This interaction can result in serious troubles affecting accelerator operations such as overheating of vacuum chamber components or instability of beam motion leading to deterioration of the beam quality or limitation of the beam intensity. Studies of collective effects are essential both for designing new accelerators and for understanding the beam dynamics in the already operated facilities in order to improve their performance.

Fundamentals of the theory of collective effects can be found in [1,2]. Interaction of relativistic charged particles with the wake fields are described using wake functions. The wake function is defined as a normalized integral of the Lorentz force that acts on a test particle moving behind a leading particle which excites the wake fields. It is a time-domain response of a vacuum chamber to a point-charge excitation. In an ultra-relativistic case, the wake function is determined only by the geometry and electromagnetic properties of the chamber, and it is independent of the beam parameters. To analyze the beam stability in most practical cases, it is enough to consider only monopole longitudinal $W_{\parallel}$ and dipole transverse $W_{\perp}$ wake functions. In the frequency domain,

each part of the vacuum chamber can be considered as a frequency-dependent longitudinal $Z_{\parallel}$ and transverse $Z_{\perp}$ impedance defined as a Fourier transform of the corresponding wake function. The beam-wake interaction intensity is proportional to the product of the impedance and beam current. The impedance of a vacuum chamber component can be approximated by few equivalent resonators with proper resonance frequencies, shunt resistances and quality factors. The beam interaction with the narrowband impedance and with the broadband one can be analyzed separately. The narrowband impedance leads to the bunch-by-bunch interaction and can result in multi-bunch instabilities, whereas the broadband impedance leads to the intra-bunch interaction and can cause single-bunch instabilities. The beam stability is analyzed using the computed impedance or its simplified representation by resonators and resistive-wall impedance calculated analytically.

For new accelerator projects, minimization of the vacuum chamber impedance is a mandatory requirement. The impedance must be minimized to avoid the chamber overheat and to provide the beam quality. There is a limited number of analytical formulas to calculate impedances of vacuum chamber sections with simple geometry, such as pill-box cavities or step transitions. In practice, components of vacuum chambers usually have more complex geometry and finite-difference simulation codes are used for computing wake functions and impedances. For an accelerator in operation, the impedance can be measured experimentally using beam-based techniques. The total frequency-dependent impedance is usually very complex so it is practically impossible to obtain its fine structure from the beam-based measurements. The measurable values are integral parameters combining

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the impedance and the bunch spectrum, such as effective impedance, longitudinal loss factor or transverse kick factor. For the longitudinal broadband impedance, the measurable effects are current-dependent bunch lengthening, synchronous phase shift, and energy spread growth due to microwave instability. For the transverse broadband impedance, the measurable effects are current-dependent shift of betatron tunes and rising/damping time of chromatic head–tail interaction.

The impedance distribution along the ring is not uniform and beam-based measurement of local impedance is a subject of importance for accelerator physics. In the case of colliders, the major contribution to the total impedance is made by few sections such as the interaction region, RF and injection sections. Measured impedance distributions of electron–positron colliders LEP and VEPP-4M are presented in [3] and [4], respectively. In the case of light sources, significant contributors are the low-gap chambers of light-generating insertion devices (IDs). A good example is published in [5], where one can see the impedance evolution of the ELETTRA storage ring from 1994 to 2002; the impedance increased after every installation of new IDs. For NSLS-II, measurement of ID impedance is a subject of practical interest because a long (4 m) low-gap (3 mm) in-vacuum undulator is proposed to be installed in the NSLS-II storage ring. Accurate estimate of its geometric and resistive-wall impedance is important to prevent beam instabilities and overheating the foil covered the undulator magnets. Precise measurement of impedances of already installed undulators together with analytical estimates and computer simulations will be very helpful.

2. Measurement technique

The interaction of a bunched beam with the transverse broadband impedance is characterized by the kick factor, an integral parameter combining the impedance and the bunch spectral density:

$$k_{\perp} = \frac{1}{2\pi} \int_{-\infty}^{\infty} Z_{\perp}(\omega) h(\omega) d\omega, \quad (1)$$

where $Z_{\perp}(\omega)$ is the frequency-dependent transverse impedance; $h(\omega) = \tilde{\lambda}(\omega) \tilde{\lambda}^*(\omega)$ is the bunch power spectrum; $\tilde{\lambda}(\omega) = \int_{-\infty}^{\infty} \lambda(t) e^{-i\omega t} dt$ is the Fourier transform of the bunch longitudinal density $\lambda(t)$ normalized as $\int_{-\infty}^{\infty} \lambda(t) dt = 1$. As it follows from the normalization, λ is expressed in s^{-1} and $\tilde{\lambda}$ is dimensionless. For a Gaussian bunch with r.m.s. length σ_t , $h(\omega) = e^{-\omega^2 \sigma_t^2}$ at low beam current. Formula (1) is valid if the bunch length is much shorter than the circumference of the machine.

A transverse dipole kick $\Delta x'$ caused by the beam-impedance interaction is

$$\Delta x' = \frac{q}{E/e} k_{\perp} x, \quad (2)$$

where $q = Ne$ is the beam charge, N is the number of particles, x is the beam transverse offset, E is the beam energy, e is the electron charge. A local transverse impedance acts on the beam as a defocusing quadrupole, strength of which depends on the beam intensity. The kicks integrated over the ring result in an intensity-dependent coherent shift of betatron tune ν_{β} . At small beam charge, the tune shift can be assumed to be linear with q and it is calculated using the formula for a small betatron detuning caused by a defocusing force:

$$\frac{\Delta \nu_{\beta}}{\Delta q} = -\frac{1}{4\pi E/e} \sum_i \beta_i k_{\perp i}, \quad (3)$$

where β_i is the beta function at the i th impedance location.

Measurement of the betatron phase advance along the ring allows determining the contributions of different sections of the vacuum chamber into the coherent shift of betatron tune. In such a way, one can obtain the azimuthal distribution of the transverse impedance [6]. Accuracy of this technique is determined by the single-turn resolution of the beam position monitors (BPMs), the signals of which are used to calculate the betatron phase. A typical coherent betatron tune shift is of the order of 0.001–0.01 per 1 mA of the beam current [7–9] and the BPM-to-BPM

phase advance is much smaller, so this technique requires very good turn-by-turn resolution of BPMs.

The orbit bump method [10,11] is more sensitive because the BPMs are used in the narrowband orbit mode rather than in the broadband turn-by-turn mode and the noise is much smaller. This method is based on the fact that an off-axis beam passing through a vacuum chamber section with non-zero transverse impedance is deflected by the wake-field kick (2) proportional to the bunch charge and its transverse position at the impedance location. So if a local orbit bump is created at the impedance location, the intensity-dependent orbit distortion is:

$$\Delta x(s) = \frac{\Delta q}{E/e} k_{\perp} x_0 \frac{\sqrt{\beta(s)\beta(s_0)}}{2 \sin \pi \nu_{\beta}} \cos(|\mu(s) - \mu(s_0)| - \pi \nu_{\beta}), \quad (4)$$

where x_0 is the orbit bump height, s_0 is the impedance location, Δq is the bunch charge variation, ν_{β} is the betatron tune, β is the beta function, and μ is the betatron phase advance. This wave-like orbit distortion can be measured using beam position monitors, and the wave amplitude is proportional to the kick factor at the bump location. To reduce the systematic error caused by intensity-dependent behavior of the BPM electronics, this error is also measured and then subtracted.

Rapid evolution of BPM electronics resulted in great improvement of the bump method sensitivity. At the very beginning of the bump method development, the orbit distortion of the order of 100 μm was measured with the accuracy of 10–40 μm [10]. The latest measurements show a possibility to measure the orbit distortion of the order of several micrometers with the sub-micron accuracy (0.2–0.5 μm) [12]. But further improvement of the measurement accuracy looks rather problematic because of the systematic errors caused by hysteresis effects of the orbit correctors and by the orbit drifts along the measurement.

To improve the bump method accuracy, we propose a technique based on excitation of an AC orbit bump by use of fast orbit correctors. The fast correctors are working in a bandwidth up to hundreds Hertz and they are typically installed at synchrotron light sources for fast orbit feedbacks. Recent developments of lattice correction techniques using an orbit response matrix measured by sine-wave excitation of fast correctors show that the orbit can be measured with a precision of the order of 0.02 μm [13,14].

To create the AC orbit bump, four fast correctors are excited by the AC driving signal

$$U_{\text{ref}}(t) = a_{\text{ref}} \sin(\omega_{\text{ref}} t + \phi_{\text{ref}}). \quad (5)$$

The same signal is used as a reference for synchronous detection of the measured beam oscillations. Since the fast correctors are designed for the fast orbit feedback, the corrector current is controlled with high stability, which is better than 10^{-4} in the NSLS-II case. Input signals of the power supply controllers are obtained by scaling U_{ref} to provide the proper currents of the four correctors pre-calculated using the lattice functions. If the bump is perfectly matched, the beam oscillates with the same frequency ω_{ref} between the outer two correctors and the orbit outside the bump is not perturbed. The bump height x_0 oscillates in time as $x_0(t) = M U_{\text{ref}}(t)$, where M is a scale factor determined by the fast corrector calibration (12.5 $\mu\text{rad/A}$ for NSLS-II) and by the lattice functions.

To measure the kick factor of a vacuum chamber section, the AC bump is created in that section and the orbit oscillation is measured by all BPMs as a function of the beam charge variation Δq . The orbit distortion (4) depending on the beam intensity oscillates in time with the same frequency as the oscillating bump height $x_0(t)$. Note that x_0 is intensity-independent, so it vanishes if we subtract the orbit x_1 of a beam with charge q_1 from the orbit x_2 of a beam with charge $q_2 = q_1 + \Delta q$. The intensity-dependent orbit distortion (4) at the location s_{bpm} of each BPM is a difference of the oscillation amplitudes measured by the BPM:

$$\Delta x(s_{\text{bpm}}) = x_2(s_{\text{bpm}}) - x_1(s_{\text{bpm}}) = a_{2\text{bpm}} - a_{1\text{bpm}}. \quad (6)$$

A standard synchronous detection technique is used for the BPM data processing, the goal of which is to extract the amplitude a_{bpm} of

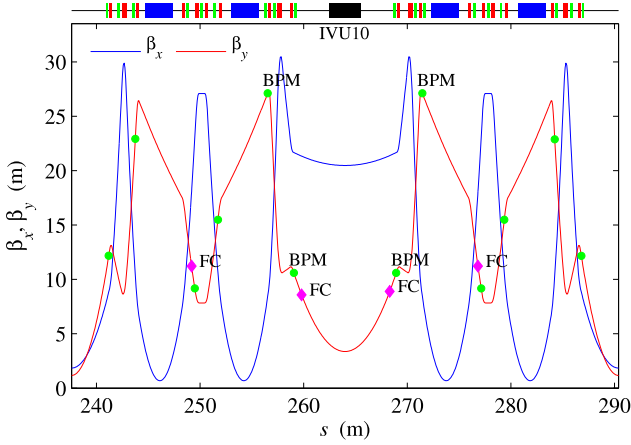


Fig. 1. Beta functions and positions of BPMs and fast correctors in the NSLS-II Cell 10. The undulator is located in the center of the section.

the measured sine-wave BPM signal. Assuming the transfer function of the entire corrector-to-BPM circuit to be linear, the BPM signal can be written as

$$U_{\text{bpm}}(t) = a_{\text{bpm}} \sin(\omega_{\text{ref}} t + \phi_{\text{bpm}}). \quad (7)$$

In the absence of noise, the product $U_{\text{mix}} = U_{\text{ref}} U_{\text{bpm}}$ is

$$U_{\text{mix}}(t) = \frac{1}{2} a_{\text{ref}} a_{\text{bpm}} \cos(\phi_{\text{bpm}} - \phi_{\text{ref}}) - \frac{1}{2} a_{\text{ref}} a_{\text{bpm}} \cos(2\omega_{\text{ref}} t + \phi_{\text{bpm}} + \phi_{\text{ref}}), \quad (8)$$

where the first term is a DC component and the second term is a double-frequency component. In the time-averaged product $\overline{U_{\text{mix}}}$, this double-frequency component vanishes if the measurement time T is equal to an integer number of the oscillation periods: $T = 2\pi k / \omega_{\text{ref}}$. Since the beam oscillation is measured by BPMs with a well-determined sampling frequency ω_s , this requirement is satisfied by choosing ω_{ref} to keep the ratio $\omega_s / \omega_{\text{ref}}$ integer. The number of measured data samples is equal to $2\pi T / \omega_s$. Then for each BPM, the oscillation amplitude a_{bpm} is obtained from $\overline{U_{\text{mix}}}$:

$$a_{\text{bpm}} = \frac{2\overline{U_{\text{mix}}}}{a_{\text{ref}} \cos(\phi_{\text{bpm}} - \phi_{\text{ref}})}. \quad (9)$$

The phase mismatch term $\cos(\phi_{\text{bpm}} - \phi_{\text{ref}})$ in the denominator is a constant factor which does not contribute to the measurement error, if it is time-independent during the measurement.

Since the fast corrector provides a highly-monochromatic sine-wave excitation to the beam, only the noise in a very narrow band around ω_{ref} contributes to the BPM signal and limits the measurement resolution. Detailed noise analysis is discussed in [14]. Sine-wave excitation of the fast orbit correctors completely eliminates the systematic error caused by hysteresis. The systematic error caused by orbit drift is also eliminated because the measured signal is not affected by the orbit motion outside the excitation frequency range.

3. Experimental results

The proposed technique has been tested at NSLS-II. To measure the vertical kick factor of a variable-gap in-vacuum undulator, four neighbor correctors are selected as shown in Fig. 1. The undulator is located in the center of the straight section, the fast correctors are marked as FC, the beam position monitors are marked as BPM, the beta functions are also shown.

The fast correctors were excited by a sine-wave drive signal with the frequency of 20 Hz. The maximal kick angle of the NSLS-II fast corrector is limited by its power supply and cannot exceed 15 μrad .

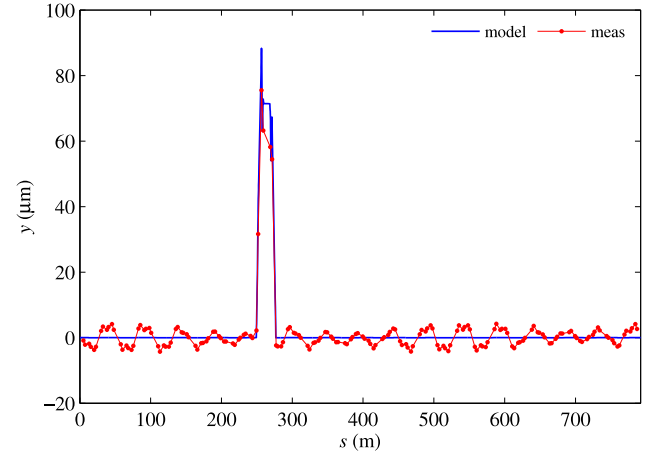


Fig. 2. Measured (red) and simulated (blue) AC orbit bumps. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The beam oscillation is measured simultaneously by 180 button-type BPMs with the sampling frequency of 10 kHz [15]. The measured noise power spectral density of NSLS-II BPMs is about 0.03 $\mu\text{m}/\sqrt{\text{Hz}}$. The BPM data were measured during $T = 10$ s, therefore the bandwidth is $\delta f = 1/T = 0.1$ Hz [14] and the noise-limited resolution is about 0.01 μm .

If these four fast correctors are excited in-phase at certain frequency with proper amplitudes, the beam oscillates with the same frequency between the outer two correctors and the orbit outside the bump is not perturbed. This is true for an ideally compensated bump. In reality, a residual orbit perturbation outside the bump is caused by the lattice model imperfections. An example of the measured AC orbit bump in comparison with the ideal (simulated) one is shown in Fig. 2. The orbit perturbation outside the bump is a source of a systematic error, but it does not exceed 5%.

The kick factor of the in-vacuum undulator has been measured in two cases: the open gap of 25 mm (minimal impedance) and the closed gap of 6 mm (maximal impedance). The vertical beta function is 3.4 m at the center of the undulator. The AC orbit was measured at two values of the beam current: 2 mA and 5 mA (the beam charge was 5.3 nC and 13.2 nC, respectively). The measured AC orbit distortions (4) are shown in Figs. 3 and 4. The error bars represent r.m.s. deviation of 10 measurements. As one can see, the measurement resolution is good enough to measure the orbit distortion of the order of 0.1 μm . The kick factor k_y is obtained by fitting the model orbit distortion (4) with the measured data using k_y as a fit parameter. The measured vertical kick factor is 478 V/(pCm) if the undulator gap is closed, and it is 152 V/(pCm) if the gap is open. Note that these values correspond to a certain bunch length. According to formulas (3) and (4), the tune shift and the orbit distortion should be proportional to the beam charge. This is true only if the kick factor is intensity-independent, that assumes a constant bunch length. This approximation is valid for small variations of the beam charge, when the intensity-dependent bunch lengthening caused by the beam interaction with broadband longitudinal impedance, is negligible. In our case, the lengthening cannot be neglected because the bunch length increased from 9.5 mm at 2 mA up to 13 mm at 5 mA. Assuming that the bunch length is proportional to the cubic root of beam intensity, and the kick factor is inversely proportional to the bunch length, the measured kick factors correspond to the bunch length of about 11 mm.

4. Conclusion

For local impedance measurement, a fast and precise technique based on excitation of an AC orbit bump using fast orbit correctors has

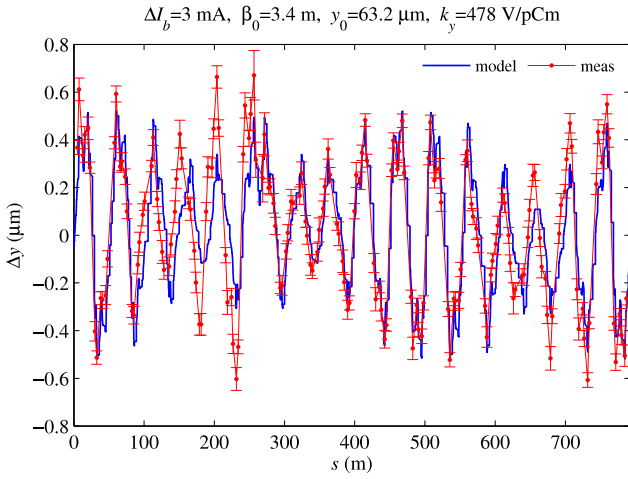


Fig. 3. Measured (red) and model (blue) AC orbit distortion. The undulator gap is 6 mm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

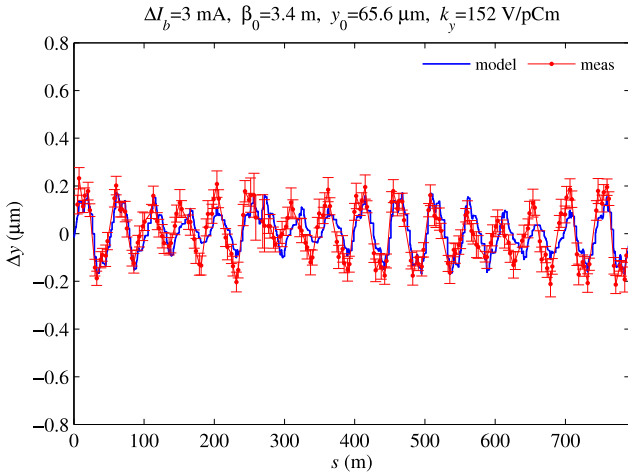


Fig. 4. Measured (red) and model (blue) AC orbit distortion. The undulator gap is 25 mm. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

been developed and tested at NSLS-II. To create the AC bump, four fast correctors adjacent to the vacuum chamber section, impedance of which is measured, are excited by an in-phase sine-wave driving signal. The synchronous detection technique is used for the BPM data processing, the narrow-band sine-wave signal allows us to improve significantly the

accuracy of the orbit bump method. Sine-wave excitation of the fast orbit correctors completely eliminates the systematic error caused by hysteresis. The systematic error caused by orbit drift is also eliminated because the measured signal is not affected by the orbit motion outside the excitation frequency range. A proof-of-principle experiment has been carried out at NSLS-II. The kick factor of one in-vacuum undulator has been measured in two cases: the open gap (minimal impedance) and the closed gap (maximal impedance). The results of this experiment have demonstrated that the measurement resolution is good enough to measure the orbit distortion of the order of 0.1 μm , which is an order of magnitude smaller than the sensitivity of the conventional bump method.

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